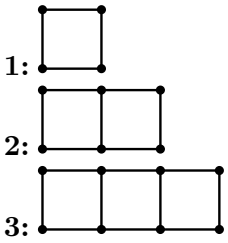
Linear Number Patterns – Extra Practice

These questions are extra practice in the same style as Questions 1 and 2 of the *Linear number patterns* worksheet: a picture pattern grows by the same amount each step, so the number of items follows a linear rule. For each question: (a) draw the next pattern, (b) complete the table, (c) find a formula, then (d)–(e) use your formula to answer questions about much larger patterns.

Question 1 – Matchstick Squares

This pattern shows the number of matchsticks needed to make a row of joined squares.



(a) Draw the next pattern (4 squares in a row).

Draw your pattern here.

(b) Copy and complete the table.

S	1	2	3	4	5	12
M						

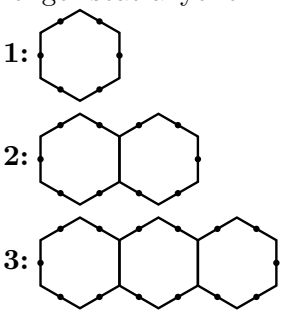
(c) Construct a formula for the number of matches (M) needed for any number of squares (S).

(d) If a row had 40 squares, how many matches would be needed?

(e) If a row had 75 squares, how many matches would be needed?

Question 2 – Hexagonal Picnic Tables

A campsite joins hexagonal picnic tables end to end in a row, as shown below (each dot is a seat). When two tables are pushed together, the shared side can no longer seat anyone.



(a) Draw the next pattern (4 hexagonal tables joined in a row).

Draw your pattern here.

(b) Copy and complete the table below.

T	1	2	3	4	5	25
C						

(c) Construct a formula for the number of seats (C) needed for any number of tables (T).

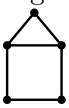
(d) If the campsite joined 40 tables in a row, how many seats would there be?

(e) If the campsite needed to seat 70 people in a single row of joined tables, how many tables would be needed?

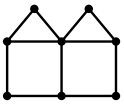
Question 3 – Matchstick Houses

This pattern shows the number of matchsticks needed to make a row of joined "houses" (a square with a triangular roof). Neighbouring houses share one wall.

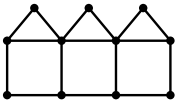
1:



2:



3:



(a) Draw the next pattern (4 houses in a row).

Draw your pattern here.

(b) Copy and complete the table.

H	1	2	3	4	5	15
M						

(c) Construct a formula for the number of matches (M) needed for any number of houses (H).

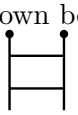
(d) If a row had 50 houses, how many matches would be needed?

(e) If a row had 80 houses, how many matches would be needed?

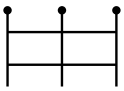
Question 4 – Garden Fencing

A fence is built from posts with two horizontal rails running between each pair of neighbouring posts, as shown below.

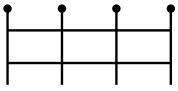
1:



2:



3:



(a) Draw the next pattern (5 posts).

Draw your pattern here.

(b) Copy and complete the table below.

P	2	3	4	5	6	7
R						

(c) Construct a formula for the number of rails (R) needed for any number of posts (P).

(d) If a fence had 50 posts, how many rails would be needed?

(e) If a fence used 200 rails, how many posts would it have?